

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech.

Branch : Civil Engineering

Semester : VII

Subject Code & Name: BTCVE705I_Bridge Engineering

Max Marks: 60

Date: 21/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	What is the typical span range of a small road bridge?	1/CO1	1				
	<table border="1"> <tr> <td>a. Less than 5 meters</td> <td>b. 5 to 15 meters</td> <td>c. 15 to 30 meters</td> <td>d. Over 30 meters</td> </tr> </table>	a. Less than 5 meters	b. 5 to 15 meters	c. 15 to 30 meters	d. Over 30 meters		
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2	The Golden Gate Bridge is an example of which type of bridge?	1/CO2	1				
	<table border="1"> <tr> <td>a. Truss</td> <td>b. Suspension</td> <td>c. Arch</td> <td>d. Cantilever</td> </tr> </table>	a. Truss	b. Suspension	c. Arch	d. Cantilever		
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3	What is the minimum vertical clearance for national highways?	1/CO2	1				
	<table border="1"> <tr> <td>a. 4.5 meters (15 feet)</td> <td>b. 5.0 meters (16.5 feet)</td> <td>c.) 5.5 meters (18 feet)</td> <td>d. 6.0 meters (19.7 feet)</td> </tr> </table>	a. 4.5 meters (15 feet)	b. 5.0 meters (16.5 feet)	c.) 5.5 meters (18 feet)	d. 6.0 meters (19.7 feet)		
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4	In the Rational Formula for calculating peak discharge (Q), which of the following is the correct formula?	1/CO3	1				
	<table border="1"> <tr> <td>a. $Q = C \cdot A \cdot I^2$</td> <td>b. $Q = C \cdot A \cdot I$</td> <td>c. $Q = C \cdot A \cdot I^{1/2}$</td> <td>d. $Q = C \cdot A \cdot I^{3/2}$</td> </tr> </table>	a. $Q = C \cdot A \cdot I^2$	b. $Q = C \cdot A \cdot I$	c. $Q = C \cdot A \cdot I^{1/2}$	d. $Q = C \cdot A \cdot I^{3/2}$		
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5	According to IRC, what is the standard classification for vehicular load in road bridges?	1/CO3	1				
	<table border="1"> <tr> <td>a. Class A, Class B</td> <td>b. Class I, Class II</td> <td>c. Class M, Class P</td> <td>d. Class AA, Class 70R</td> </tr> </table>	a. Class A, Class B	b. Class I, Class II	c. Class M, Class P	d. Class AA, Class 70R		
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6	Longitudinal forces are essential to consider when designing for:	1/CO1	1				
	<table border="1"> <tr> <td>a. Earthquakes</td> <td>b. Dynamic load impact</td> <td>c. Temperature-induced expansion</td> <td>d. All of the above</td> </tr> </table>	a. Earthquakes	b. Dynamic load impact	c. Temperature-induced expansion	d. All of the above		
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7	Which design code is primarily used for highway and road bridge design in India?	1/CO3	1				
	<table border="1"> <tr> <td>a. AASHTO</td> <td>b. BS Code</td> <td>c. IRC Code</td> <td>d. Eurocode</td> </tr> </table>	a. AASHTO	b. BS Code	c. IRC Code	d. Eurocode		
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8	Which of the following is NOT a type of expansion joint used in bridge design?				1/CO2	1
	a. Finger Joint	b. Modular Joint	c. Hinged Joint	d. Asphalt Expansion Joint		
9	Which of the following design methodologies is considered preferable for the analysis of bridge superstructures?				2/CO3	1
	a. Orthotropic Plate Theory	b. Grillage Analysis	c. Finite Element Analysis	d. Load Distribution Techniques		
10	Which of the following is an important parameter in the selection of a bridge superstructure?				2/CO2	1
	a. Soil Type	b. Length and Span of the Bridge	c. Traffic Load	d. All of the above		
11	What is the primary function of wing walls in a bridge?				2/CO1	1
	a. To provide stability to the bridge superstructure	b. To support the bridge deck	c. To protect the abutments from soil erosion	d. To support expansion joints		
12	Which type of foundation is most suitable for a bridge in a riverbed?				2/CO1	1
	a. Open Foundation	b. Well Foundation	c. Pile Foundation	d. Slab Foundation		
Q. 2 Solve the following.						12
A)	Explain the main components of a road bridge with the help of neat sketch and describe the function of each component in supporting the bridge structure.				2/CO1	6
B)	Mention the ways in which bridges are classified.				1/CO1	6
Q.3 Solve the following.						12
A)	What are the environmental considerations that need to be addressed when collecting data for the design of a river bridge?				2/CO2	6

B)	A bridge is proposed to be constructed across a river with a discharge of $400 \text{ m}^3/\text{s}$. Assuming the silt factor $f=1.20$, determine the maximum scour depth when the bridge consists of 4 spans, each of 25 m (Assume $R = 2$). Also discuss the factors that influence the scour depth around bridge piers.	3/CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Explain the different types of live load classifications for road bridges as per IRC-6. How does the classification system help in determining the design loads for various types of vehicles?	2/CO2	6
B)	Explain the different types of loads that must be considered in the design of road bridges according to the IRC, BS, and AASHTO codes. Discuss the significance of dead load and live load in bridge design.	3/CO2	6
C)	What factors should be considered in determining the appropriate sidewalk width for urban and rural roads?	2/CO2	6
Q.5	Solve Any Two of the following.		12
A)	Discuss the advantages of using Finite Element Analysis (FEA) in the design of bridge superstructures. How does FEA provide more accurate results compared to traditional methods?	3/CO3	6
B)	Discuss the different types of bridge bearings and their functions. How are they designed to accommodate movements due to temperature, traffic load, and settlement?	3/CO4	6
C)	What are the different types of expansion joints used in bridge design? Explain how they are designed to accommodate movement and prevent structural damage.	2/CO3	6
Q. 6	Solve Any Two of the following.		12
A)	Define following in detail : i) Abutment ii) Pier iii) Wing wall	1/CO1	6
B)	Differentiate between open foundations, pile foundations, and well foundations used in bridge construction. Provide examples of bridges where each foundation type is commonly used.	3/CO2	6
C)	Discuss the importance of soil-structure interaction in the design of bridge substructures. How does it influence the performance and safety of the bridge?	3/CO2	6
	*** End ***		